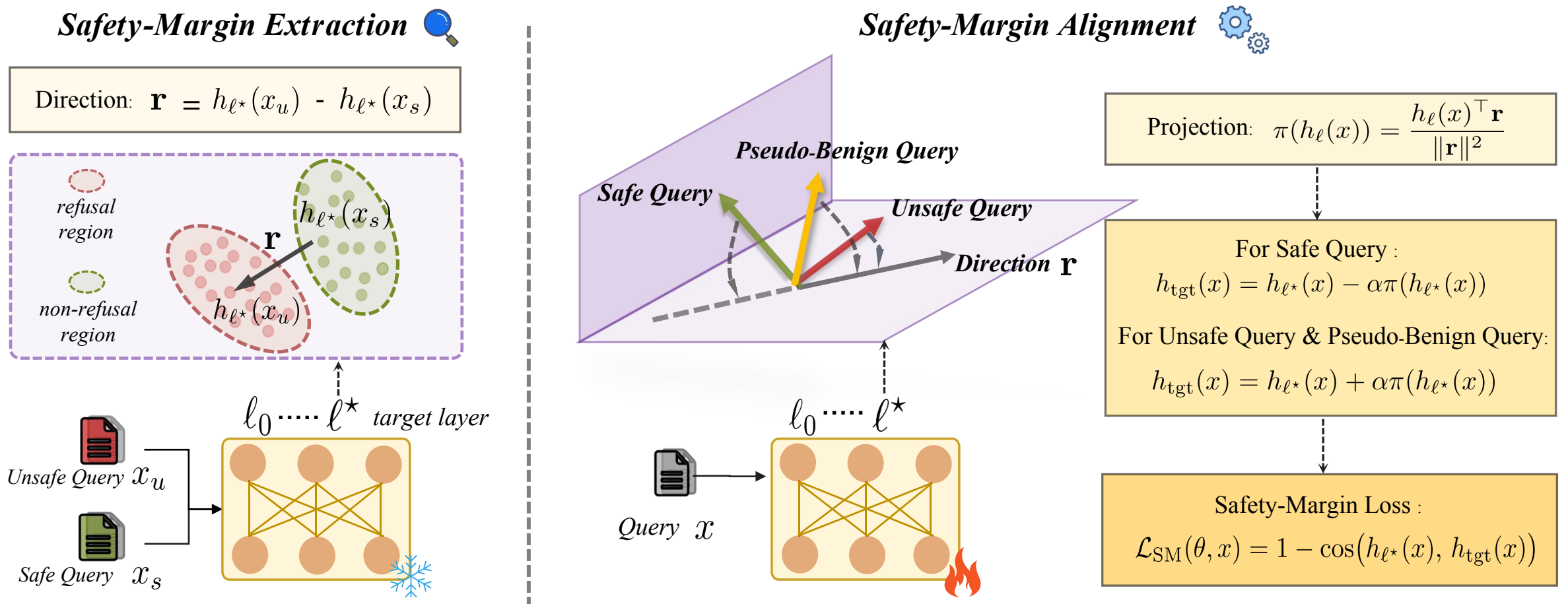
Safety-Margin Extraction
Direction: $\mathbf{r} = h_{\ell^\star}(x_u) - h_{\ell^\star}(x_s)$
refusal region
non-refusal region
$h_{\ell^\star}(x_s)$
$\mathbf{r}$
$h_{\ell^\star}(x_u)$
$\ell_0 \cdots\cdots \ell^\star$ target layer
Unsafe Query x_u
Safe Query x_s
Safety-Margin Alignment
Pseudo-Benign Query
Safe Query
Unsafe Query
Direction $\mathbf{r}$
$\ell_0 \cdots\cdots \ell^\star$
Query x
Projection: $\pi(h_\ell(x)) = \frac{h_\ell(x)^\top \mathbf{r}}{\|\mathbf{r}\|^2}$
For Safe Query :
$h_{\text{tgt}}(x) = h_{\ell^\star}(x) - \alpha\pi(h_{\ell^\star}(x))$
For Unsafe Query & Pseudo-Benign Query:
$h_{\text{tgt}}(x) = h_{\ell^\star}(x) + \alpha\pi(h_{\ell^\star}(x))$
Safety-Margin Loss :
$\mathcal{L}_{\text{SM}}(\theta, x) = 1 - \cos\big(h_{\ell^\star}(x),\, h_{\text{tgt}}(x)\big)$